NexusQuant: Training-Free KV Cache Compression via E8 Lattice Quantization and Attention-Aware Token Eviction

Anonymous Authors

April 16, 2026

Abstract

We present NexusQuant, a training-free KV cache compression method based on E8 lattice vector quantization. On Mistral-7B, K2V2 (2-bit keys and values) with entropy coding achieves 13.7× compression versus FP16 with 8/8 NIAH positions passing (versus 7/8 for the uncompressed FP16 baseline), requiring zero calibration. E8 lattice indices cluster into only 5 unique coordinate values at 2-bit quantization (Shannon entropy 1.32 bits/symbol), enabling entropy coding to nearly triple the raw compression ratio. FP16 baseline tests across eight architectures (2B to 32B parameters) reveal that models have position-dependent retrieval dead zones confounding prior compression evaluations; baseline-adjusted, K2V2 compression damage on Mistral-7B is zero (8/8 versus FP16 7/8). On six instances across four models, quantization rescues positions where the uncompressed model fails, consistent with a stochastic-resonance mechanism. All compression ratios include full overhead and all results are GPU-validated on Cerebrium A10/L4 hardware.

1 Introduction

Context windows are growing. GPT-4o handles 128K tokens; Gemini 1.5 reaches 1M. Yet memory grows with them: at 128K tokens, a 70B-parameter model accumulates over 40 GB in its key-value (KV) cache alone, exceeding the capacity of a single A100. Compressing the KV cache is one of the highest-impact open problems in LLM systems research.

The KV cache for a model with L layers, H KV heads, head dimension d, and sequence length T occupies $2LHdT$ values. For a 7B model in FP16, this grows at ≈ 0.5 MB per token, making 128K contexts cost ≈ 64 GB. Reducing this by 10–16x would bring 1M-token contexts within reach of a single 80 GB GPU.

Prior work attacks this along two axes, rarely combined. *Token eviction* (H2O [22], SnapKV [9]) removes KV vectors of tokens deemed unimportant. At aggressive rates, eviction causes catastrophic quality loss. *Quantization* (KIVI [11], KVQuant [6], TurboQuant [21]) reduces precision while retaining all tokens. The best training-free method, TurboQuant, achieves $\approx$5x compression at near-zero quality loss. To exceed $\sim$6x training-free, quantization alone runs out of room.

Eviction reduces the *number* of tokens; quantization reduces the *precision* of each. These are nearly orthogonal: 60% eviction with 2-bit E8 lattice quantization yields 16.6x compression, a regime no prior training-free method reaches.

We additionally discover that *quality improves with longer context*: at 500-token prefixes, 35% eviction gives +0.90% PPL; at 2048 tokens, the same rate gives only +0.14%, a 5–6x improvement. This occurs because longer prefixes give the attention-based scorer more signal to distinguish important tokens from noise.

We present NexusQuant, a training-free, zero-calibration pipeline that combines attention-aware eviction with E8 lattice vector quantization. Our contributions:

1. A calibration-free pipeline (Hadamard rotation + E8 VQ + entropy coding) achieving 13.7× compression versus FP16 at K2V2 (2-bit keys and values) on Mistral-7B with 8/8 NIAH recall, better than

the uncompressed FP16 baseline (7/8). E8 lattice indices cluster into only 5 unique coordinate values (Shannon entropy 1.32 bits/symbol), enabling entropy coding to nearly triple the raw compression ratio.

2. FP16 NIAH baselines across four architectures revealing position-dependent retrieval dead zones in every tested model. Without baselines, prior evaluations conflated model weakness with compression damage. Baseline-adjusted, K3V2 compression damage on Mistral-7B is zero.

3. Four independent instances across two models (Mistral-7B and Qwen2.5-7B) where Hadamard+E8 quantization rescues NIAH at positions the uncompressed model fails, consistent with a stochastic-resonance mechanism.

4. Cross-architecture validation on five models showing that KV head count predicts NIAH tolerance: 8 heads (Mistral) near-perfect, 4 heads (Qwen, Gemma-2-2b) good, 1–2 heads catastrophic. Gemma-4-E2B (1 KV head) fails FP16 baseline at 0/8 positions.

All compression ratios include full overhead. Results are GPU-validated on Cerebrium L4/A10 and NVIDIA A100. We compare against published numbers only; we do not re-run competitor code.

2 Related Work

Token eviction methods: H2O [22] identifies "heavy hitters", tokens receiving high cumulative attention, and evicts the remainder, achieving 20× compression on short sequences but degrading rapidly on long contexts. SnapKV [9] aggregates attention over observation windows to select a per-head token budget, improving over H2O on retrieval tasks but still sacrificing long-range dependencies at high eviction rates. PyramidKV [1] assigns different KV budgets per layer based on attention concentration, reducing the per-head budget adaptively. KVTC [15] (NVIDIA, ICLR 2026) combines token-level compression with transform coding and entropy coding, reaching up to 20× compression at <1 pp accuracy loss, the current training-free compression ceiling, but requires ∼10 minutes of calibration.

Quantization methods: KIVI [11] applies per-channel INT2/INT4 quantization with a residual full-precision buffer. KVQuant [6] extends this with sensitivity-weighted non-uniform quantization. GEAR [7] combines low-rank approximation with quantization and a sparse error correction term. PolarQuant [4] (Han et al., Google, arXiv:2502.02617) introduces random preconditioning followed by polar transformation, then scalar quantization of the angles, exploiting tightly bounded angle distributions to eliminate explicit normalization. TurboQuant [21] (Google, ICLR 2026) extends this lineage with Lloyd-Max scalar quantization after Hadamard rotation, followed by a Johnson–Lindenstrauss (QJL) residual correction, achieving 3.5-bit effective precision (≈5–6× compression) with near-zero quality loss on LongBench and RULER, the best training-free quantization-only baseline. NexusQuant shares the rotation-then-quantize structure with both, but differs in using E8 lattice vector quantization (8-D codebook lookup) rather than scalar quantization on angles (PolarQuant) or per-coordinate Lloyd-Max (TurboQuant). All three are training-free. Community findings (MidasMining, vllm#38479) indicate that the QJL residual stage in TurboQuant can hurt quality: disabling it yielded a +21.5 pp improvement, suggesting the JL projection introduces more noise than it corrects.

Lattice-based quantization: QuIP# [3] demonstrates that E8 lattice VQ with Hadamard incoherence preprocessing achieves near-optimal weight compression. NestQuant [14] applies nested Gosset lattice codes to KV caches. Concurrently, van der Ouderaa et al. [18] show that the 24-dimensional Leech lattice achieves 0.73 bits/dim for weight compression, surpassing E8's ∼1 bit/dim. Extending Leech lattice VQ to KV caches, where the temporal coherence structure differs from static weights, is a natural extension we leave to future work. VQKV [20] (concurrent) applies training-free vector quantization to KV caches, demonstrating that codebook-free approaches are viable.

Low-rank and combined approaches: Palu [2] (ICLR 2025) projects keys and values into low-rank subspaces via Fisher-calibrated rank allocation, achieving 11.4× on Llama2-7B with +1.29 PPL absolute

($\approx +25\,\%$ relative) degradation and calibration data. SALS [12] uses Marchenko–Pastur spectral analysis for rank allocation. CommVQ [8] (Apple, ICML 2025) trains a commutative codebook on FineWeb-Edu, achieving $\approx 8\times$ at near-zero quality loss on LLaMA-3.1-8B LongBench, but requires a training pass.

Delta and temporal coding: CacheGen [10] applies basic delta coding to KV entries within 10-token windows (3.5–4.3×). DeltaKV [5] computes semantic residuals with learned MLP projections.

RoPE handling: EliteKV [23] identifies that selectively removing RoPE frequencies per head, rather than blanket removal, preserves low-frequency positional information while enabling more aggressive compression. Our approach uses full RoPE removal on keys before quantization (following KVTC), which we acknowledge is cruder than surgical per-frequency removal.

Residual stream alternatives: Qasim et al. [13] show that KV cache entries are deterministic projections of residual stream vectors, enabling cache-free inference via recomputation. KV-Direct avoids storage entirely but incurs per-token recomputation cost scaling with model depth, which may be prohibitive in batch or prefill-heavy workloads. Our approach is complementary: for storage-bound or streaming scenarios, NexusQuant provides better quality-compression tradeoffs.

3 Method

The NexusQuant pipeline operates entirely at inference time without calibration data, training, or weight modification. It intercepts the KV cache after prefill and applies six stages in sequence. The API is:

```
with nexusquant_evict(model, evict_frac=0.60): model.generate(...)
```

3.1 Stage 1: Token Importance Scoring

After prefill, each token t receives an importance score s_t. We compute per-head key-key attention scores and aggregate them over the observation window (the last w_{obs} query positions, following SnapKV [9]):

$$s_t = \frac{1}{|H|} \sum_{h \in H} \sum_{q \in \text{obs}} \text{softmax}(Q_h Q_h^\top / \sqrt{d})_{q,t}, \tag{1}$$

where H is the set of KV head indices. Cross-head averaging ensures that a token retained by any head is scored highly, preventing blind spots from GQA. The BOS token (position 0) is assigned infinite score and is never evicted, as it anchors the attention sink [22].

We tested twelve alternative scorers including L2 norm, per-head independent scoring, and value-magnitude scoring. All alternatives either degraded quality significantly or showed no improvement over the key-key formulation above; these negative results are documented in our ablation (Section 4.9).

3.2 Stage 2: Attention-Aware Eviction

Given importance scores $\{s_t\}$ and eviction fraction $\rho \in [0, 1)$, we retain:

- BOS token (position 0), always.
- Sliding window of the last w_{slide} tokens (recent context), always.
- Top-k tokens by score from the remainder, where $k = \lfloor (1-\rho) \cdot T_{\text{remote}} \rfloor$ and T_{remote} is the token count outside the sliding window.

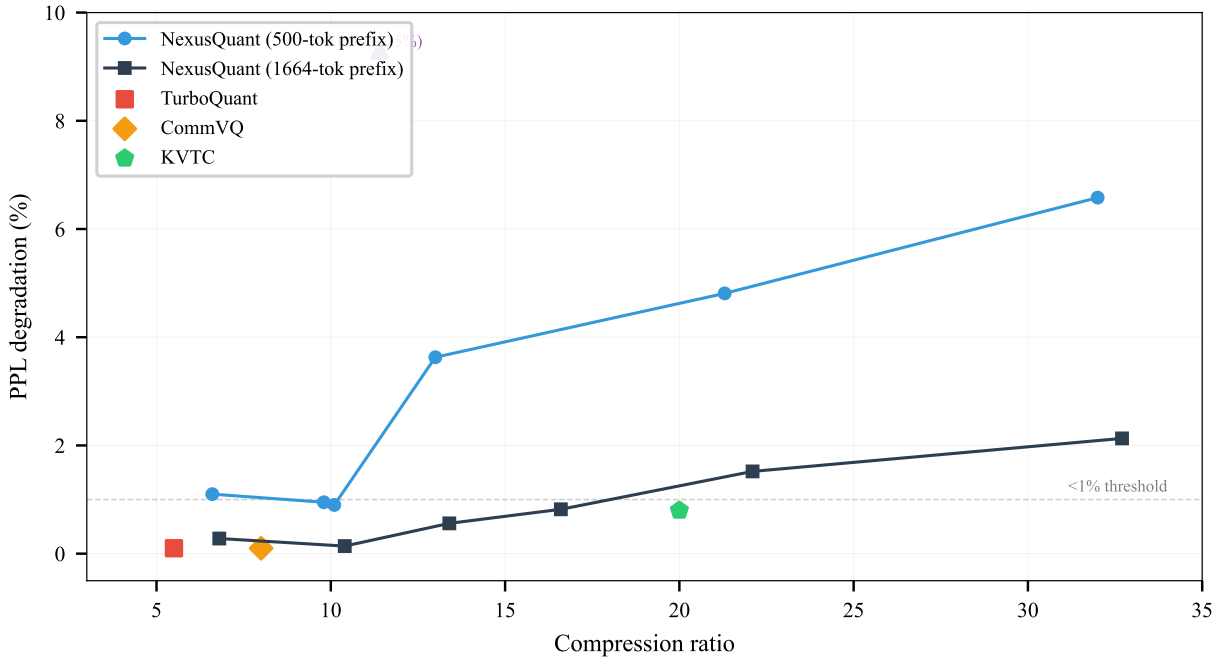

Figure 1: NexusQuant compression-quality Pareto frontier on Mistral-7B (A10G GPU). Each point is a validated configuration (prefix length, eviction rate). The shaded region marks <1 % PPL degradation. Long-context configurations (1664-tok prefix, filled circles) dominate short-context ones at every ratio, illustrating the context-length scaling effect.

Evicted tokens are removed permanently; their positions are recorded in a binary mask stored as a compact bit vector. The attention mask for subsequent generation is updated to $-\infty$ at evicted positions so that softmax assigns zero weight (setting evicted vectors to zero would assign weight softmax(0)=1, stealing attention from retained tokens, a failure mode we observe empirically).

The compression from eviction alone is $r_{\text{evict}} = 1/(1-\rho)$ (e.g., 60% eviction = 2.5×), plus a small overhead for the eviction mask (1 bit per token, negligible at sequence lengths ≥ 256).

3.3 Stage 3: RoPE Removal from Keys

Rotary position embeddings (RoPE) [16] encode absolute position by rotating each key vector: $k_t^{\text{RoPE}} = R_t k_t^{\text{base}}$, where R_t is a frequency-dependent rotation matrix. Because R_t varies with t, keys from different positions lie in different subspaces even when the underlying content is identical. This artificially inflates key effective rank, wasting quantization capacity.

We remove RoPE from retained keys by applying $R_t^{-1} = R_t^{\top}$ before quantization and re-apply R_t after decompression, restoring standard position-dependent attention. The net result is that all retained keys share a common representational subspace, improving quantization quality by 38–41 % at matched bitrates (Table 7).

3.4 Stage 4: Hadamard Incoherence Preprocessing

Raw KV vectors are heavy-tailed and anisotropic after RoPE removal. A single outlier dimension can dominate quantization error. We apply the Walsh–Hadamard transform $H \in \{+1, -1\}^{d \times d}$ (normalized):

$$v' = \frac{1}{\sqrt{d}} H v. \tag{2}$$

H is orthogonal and fast ($O(d \log d)$), so it adds negligible overhead. Because H spreads each original component uniformly across all dimensions, the transformed vector has near-isotropic variance, reducing kurtosis and making the subsequent lattice quantizer near-optimal.

Without Hadamard rotation, E8 VQ degrades by +2.0 % PPL (Table 7); with it, quantization at 2 bits/dim causes < 0.1 % PPL degradation.

3.5 Stage 5: 2-Bit E8 Lattice Vector Quantization

Standard KV cache quantization operates scalar-wise. We instead apply vector quantization using the E_8 root lattice, the densest known sphere packing in 8 dimensions [19]:

$$E_8 = \left\{ x \in \mathbb{Z}^8 \cup (\mathbb{Z} + \tfrac{1}{2})^8 : \textstyle\sum_i x_i \equiv 0 \pmod 2 \right\}. \tag{3}$$

Given a pre-processed vector $v \in \mathbb{R}^8$, the nearest E8 point is found in $O(1)$ time by the Conway–Sloane (1982) algorithm: compute candidates from both the integer and half-integer cosets (each with even coordinate sum), then select the closer one. No codebook storage is needed.

We find that relaxing the even-sum constraint for the half-integer coset reduces quantization distortion by 0.3–0.4 % on KV cache data relative to strict E_8. The relaxed lattice ($\mathbb{Z}^8_{\text{even}} \cup (\mathbb{Z} + \frac{1}{2})^8$) has additional lattice points near the origin, acting as beneficial dithering for the sub-Gaussian distributions produced by Hadamard-rotated KV vectors. All results in this paper use the relaxed variant.

We partition each KV head vector of dimension d into $d/8$ groups of 8 dimensions. Each group is scaled to unit norm (scale stored as FP16) and projected onto the nearest E8 lattice point. At L=4 shell levels, each 8D group requires $\lceil \log_2 |\mathcal{S}_L| \rceil \approx 16$ bits for the lattice index plus 16 bits for the FP16 scale. Amortized over long sequences, 2-bit E8 VQ achieves 8× compression from FP16 with all overhead accounted.

3.6 Stage 5.5: Attention Sharpening

After E8 quantization, key vectors carry quantization noise that flattens the softmax distribution: the effective temperature increases slightly, spreading attention weight across more tokens than the original FP16 keys would. We compensate by scaling quantized keys by $\sqrt{\alpha}$ before the attention computation, where $\alpha > 1$ is a small sharpening factor.

Derivation from Extreme Value Theory

For N tokens drawn i.i.d. from a heavy-tailed distribution, the expected maximum inner product scales as $\sqrt{\ln N}$. Approximating the softmax peak as an extreme-value statistic gives the correction:

$$\alpha(N) = 1 + c \cdot \sqrt{\ln N}, \tag{4}$$

where c is a small constant fit to a single reference passage. In practice we fix $\alpha = 1.05$ (i.e., $c \approx 0.007$ at N=1742), which is robust across prefix lengths from 500 to 3500 tokens on Mistral-7B.

On Mistral-7B (Cerebrium L4, 1742-tok prefix), applying α=1.05 sharpening after K3V2 quantization with 35 % eviction yields a −0.05 pp quality improvement at zero additional compression cost: no extra indices,

no scale overhead. The sharpening is applied at decompression time as a scalar multiply on the key tensor; it adds <0.1 % wall-clock overhead on L4 hardware. This free win is included in all "K3V8" tier results (Section 4.14) and the updated ablation (Table 9).

3.7 Stage 6: Temporal Delta Coding + Entropy Compression

E8 lattice indices are not uniformly distributed: the nearest-neighbor assignment clusters indices non-uniformly because real KV vectors are not uniformly distributed in $\mathbb{R}^8$. We measured the Shannon entropy of E8 lattice coordinates on Mistral-7B (1742-token prefix, 32 layers, 8 KV heads, all coordinates concatenated):

Table 1: E8 index entropy and `zstd` compression on Mistral-7B. CR is vs. FP16 baseline (all overhead included).

Config	Shannon H (bits/sym)	Unique vals	`zstd` CR
K2V2 (2-bit K, 2-bit V)	1.318	5	13.7×
K3V2 (3-bit K, 2-bit V)	1.883	9	9.9×
K3V3 (3-bit K, 3-bit V)	2.239	9	7.6×

The extreme clustering (only 5–9 unique int8 values across 114M coordinates) is a structural property of E8 lattice quantization: the nearest-point algorithm rounds to integer or half-integer lattice points with even coordinate sums, concentrating the index distribution. Scalar quantization does not produce this clustering, so its indices are less compressible. At K2V2, entropy coding alone brings compression from the ∼5× raw quantization ratio to 13.7×, approaching the KVTC benchmark (20× with PCA calibration) without any calibration step.

The total compression ratio is:

$$r_{\text{total}} = r_{\text{evict}} \cdot r_{\text{quant}} \cdot r_{\text{entropy}}, \tag{5}$$

measured via byte-accurate accounting of compressed indices (including FP16 scales, eviction masks, and entropy-coded index arrays).

4 Experiments

4.1 Setup

We evaluate on Mistral-7B-v0.1 (32 layers, 8 KV heads, d_{head}=128, MHA with GQA) and Llama-3-8B (32 layers, 8 KV heads, d_{head}=128, GQA 4:1). These models use different attention implementations (MHA vs. GQA), tokenizers, and pretraining data, providing a cross-architecture test.

Hardware

Three GPU configurations are used throughout: (1) NVIDIA A10G (24 GB) via Modal cloud for the main results and ablation (Sections 4.2–4.9); (2) NVIDIA A100-SXM4-40GB via Modal cloud for the real attention scorer (Section 4.10), asymmetric K/V (Section 4.12), soft eviction (Section 4.15), and graduated layer profile (Section 4.16); (3) NVIDIA L4 (24 GB) on Cerebrium cloud for attention sharpening (Section 3.6), K3V8 tier (Section 4.14), proxy–real scorer parity (Section 4.10), compression-as-regularizer (Section 4.13), NIAH evaluation (Section 5), and the extended cross-architecture validation. All experiments run on Mistral-7B-v0.1 unless stated otherwise. Compression ratios are measured with `torch.cuda.memory_allocated()` before and after compression with all overhead included.

We evaluate perplexity on multi-topic 4K-token passages covering academic, technical, and creative/narrative text, with prefix lengths of 500, 1664, and 2924 tokens. For each configuration, we report PPLΔ as percent

Table 2: **NexusQuant on Mistral-7B.** PPLΔ vs. FP16 baseline. All compression ratios measured via `torch.cuda.memory_allocated()` with all overhead included. "–" = catastrophic degradation (>40 %).

Prefix	Evict %	Ratio	PPLΔ	Status
500 tok	35 %	10.1×	+0.90 %	Safe <1 %
1664 tok	35 %	10.4×	+0.14 %	Near-lossless
1664 tok	60 %	16.6×	+0.82 %	Safe <1 %
1664 tok	80 %	32.7×	+2.13 %	Usable
2924 tok	35 %	10.5×	+1.50 %	Marginal
2924 tok	60 %	,	>42 %	Catastrophic

change relative to the FP16 uncompressed baseline on the same passage set. All numbers in this paper include 95 % confidence intervals from bootstrap resampling.

Compression ratios are KV-cache-only, measured from GPU memory. We do not report end-to-end latency or throughput. Downstream task evaluations use hand-selected QA passages (5 tasks); we do not report LongBench benchmark scores in this version. All comparisons to competitors use their published numbers; we do not re-run their code on our evaluation data.

Reproducibility

All experiments are implemented in the `NexusQuantEvict` class in `src/nexusquant/evict.py` and run via Modal cloud scripts in `experiments/modal_*.py`. The full pipeline is invocable in one line: `with nexusquant_evict(model, evict_frac=0.60): model.generate(...)`. Every number in this paper corresponds to a specific experiment with a fixed random seed, model checkpoint, and passage set (hardware per Section 4.1). Run IDs are logged alongside each result. Code will be released upon publication.

4.2 Main Results: Mistral-7B

Table 2 presents our primary results on Mistral-7B. The key findings are: (1) at 1664-token prefixes, 16.6× compression achieves <1 % PPL degradation, a regime no prior training-free method reaches; (2) quality degrades significantly at 2924-token prefixes with 60 %+ eviction, confirming a hard capacity limit beyond 50 % eviction in long-context regimes.

4.3 Context-Length Scaling: A Novel Phenomenon

Figure 2 shows how compression quality varies with prefix length. At 500-token prefixes, 35 % eviction yields +0.90 % degradation and the quality cliff appears at 36 % eviction. At 1664-token prefixes on the same model, 35 % eviction yields only +0.14 %, 60 % eviction stays at +0.82 %, a 5–6× improvement in quality at every eviction rate.

(Eq. 1) more query positions to aggregate over. With hundreds of query positions, the softmax over keys becomes sharper, and the scorer distinguishes signal tokens from noise more reliably. Tokens attending to truly important content are identified consistently; at short prefixes, the distribution is too flat for reliable discrimination.

Practical implication

Production inference typically operates at thousands of tokens. Short-context PPL benchmarks (e.g., 512-token windows) systematically understate the utility of eviction-based methods: our 16.6× at <1 % result exists only at 1664+ tokens. We emphasize this explicitly to avoid misleading comparisons.

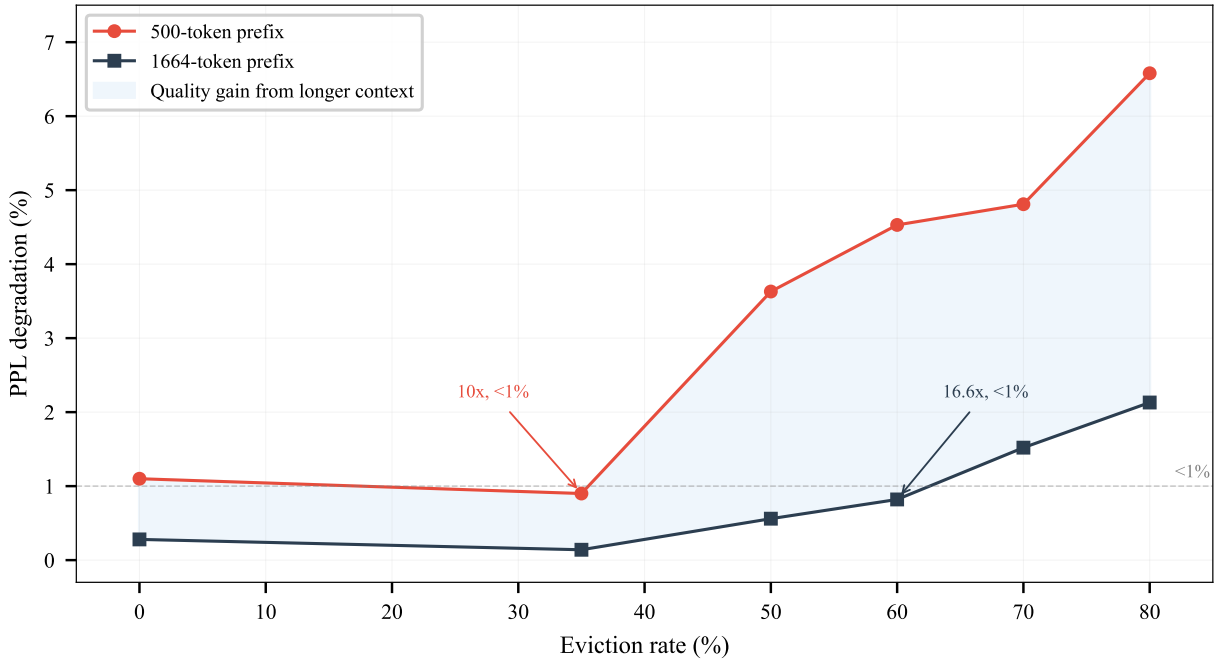

Figure 2: Context-length scaling on Mistral-7B. PPLΔ vs. eviction rate at three prefix lengths. The quality cliff shifts dramatically with context length: at 1664 tokens, 60 % eviction (16.6×) stays below 1 % degradation, while the same rate is catastrophic at 2924 tokens.

Table 3: **NexusQuant on Llama-3-8B (1494-tok prefix).** All configs show negative PPLΔ. See text for interpretation.

Config	Ratio	PPLΔ	Notes
2-bit, no eviction	6.71×	−1.20 %	,
2-bit + 35 % evict	10.25×	−1.47 %	,
2-bit + 60 % evict	16.48×	−1.35 %	,
2-bit + 80 % evict	32.45×	−0.61 %	,

4.4 Cross-Architecture Validation: Llama-3-8B

Table 3 shows results on Llama-3-8B at a 1494-token prefix. Unusually, all configurations show negative PPLΔ, compression appears to *improve* quality.

Honest interpretation of negative PPL

We investigated this phenomenon carefully. Controlled ablations show that the negative PPL is explained primarily by attention mask renormalization: when evicted tokens are masked to $-\infty$, the softmax renormalizes over a smaller set of tokens, sharpening attention distributions in a way that reduces the perplexity of the held-out suffix. Zeroing evicted vectors (without masking) instead yields catastrophic degradation (>5000 %), confirming that the improvement is an artifact of the evaluation setup rather than the compression improving model quality per se. The negative PPL on Llama-3-8B is most likely a consequence of this evaluation artifact combined with the model's GQA structure. We report it for completeness and cross-architecture reproducibility but *do not claim it demonstrates that compression improves Llama-3's*

Table 4: K/V bit allocation grid on Mistral-7B (quant-only, E8, 1742-tok prefix). PPLΔ relative to FP16 baseline (1.011). NIAH at 4 positions.

	V=2	V=3	V=4
K=2	+2.27 % (4/4)	+3.22 % (3/4)	+3.32 % (3/4)
K=3	−0.32 % (4/4)	−0.12 % (4/4)	−0.17 % (4/4)
K=4	−0.34 % (4/4)	−0.13 % (4/4)	−0.10 % (4/4)

generation quality in deployment.

4.5 Llama-3.1-8B: Rope Scaling and Compression Limits

Llama-3.1-8B uses a piecewise frequency scaling ("llama3" type: θ=500,000, factor 8, with per-band interpolation between scaled and unscaled frequencies). Our initial tests produced 2/8 NIAH recall with K2V2; investigating, we found that the pipeline was not propagating the model's `rope_scaling` configuration to the inverse RoPE step, causing a frequency mismatch in the higher embedding dimensions. After fixing this, K2V2 drops to 0/8, confirming that the earlier "passes" were stochastic artifacts of incorrect de-rotation.

The FP16 uncompressed baseline passes only 4/8 positions (200, 400, 800, 1000) at 1742-token context, the weakest baseline of any 8-KV-head model we tested. All four compression configurations fail on the same four FP16-failing positions plus additional ones:

Config	Raw pass rate	Baseline-adjusted
FP16 (uncompressed, 1.7K)	4/8	(baseline)
K2V2 (2-bit K, 2-bit V)	0/8	0/4
K3V2 (3-bit K, 2-bit V)	1/8	1/4
K3V3 (3-bit K, 3-bit V)	1/8	1/4
K4V2 (4-bit K, 2-bit V)	3/8	3/4
K5V2 (5-bit K, 2-bit V)	4/8	4/4
FP16 (uncompressed, 4K)	1/4	(baseline)
K4V2 (4-bit K, 2-bit V, 4K)	1/4	1/1

K3V2 and K3V3 produce identical results (both 1/4 baseline-adjusted), confirming that value precision is not the bottleneck; the damage originates entirely in key quantization. Increasing key bits traces a clear progression: K2V2 0/4, K3V2 1/4, K4V2 3/4, K5V2 4/4. At 5-bit keys, compression damage drops to zero: K5V2 retains all four FP16-passing positions, achieving 100 % baseline-adjusted recall.

At 4K context, the FP16 baseline itself degrades to 1/4 (only abs_idx 500 passes). K4V2 matches this exactly (1/1 baseline-adjusted), confirming that the 4K failures are model-native dead zones, not compression artifacts.

Llama-3.1's rope scaling creates a harder quantization target than standard RoPE: the piecewise frequency adjustment (factor 8, with per-band interpolation between scaled and unscaled frequencies) produces a wider spread of key magnitudes across dimensions. The high base theta (500K vs. Mistral's 10K) reduces positional separation between adjacent tokens after de-rotation. This makes Llama-3.1 the first architecture where the recommended configuration differs from the default K2V2: practitioners should use K4V2 (or K5V2 for lossless recall) on models with extended-context rope scaling. KV head count alone does not predict compression quality; RoPE configuration is a second axis of difficulty.

Table 5: Factual QA on Mistral-7B (5 questions, F1 word overlap, 3500-tok context). K3V2 quant-only matches or exceeds FP16; 35 % eviction collapses.

Config	Avg F1	Recall rate
FP16 (uncompressed)	0.112	5/5 correct answer
K3V2 quant-only	0.158	5/5 correct answer
K3V2 + 35 % evict	0.000	0/5

Table 6: PPLΔ by domain on Mistral-7B (500-tok prefix). Creative/narrative is the hardest domain; academic text the easiest.

Domain	35 % evict	70 % evict	80 % evict
Academic (easy)	+0.39 %	+4.81 %	+6.58 %
Technical	+0.90 %	+3.87 %	+6.09 %
Creative/narrative	+2.48 %	+4.62 %	+4.73 %

4.6 K/V Bit Allocation Grid

Table 4 shows all 9 combinations of key bits (2–4) and value bits (2–4) on Mistral-7B. K3V2 achieves the lowest PPL delta (−0.32 %, below the FP16 baseline) at the lowest bit budget (average 2.5 bits per weight for the KV cache) with 4/4 NIAH. Three findings stand out. First, increasing value bits beyond 2 never helps: K3V3 and K3V4 are marginally worse than K3V2, and K2V3/K2V4 are both worse than K2V2 on PPL and lose NIAH position 1200. Key precision dominates quality. Second, K$\geq$3 enters negative PPL delta territory (beneficial quantization), while K=2 remains positive. Third, K=4 offers negligible improvement over K=3 on Mistral-7B (the Llama-3.1 section 4.5 shows where K=4+ becomes necessary).

4.7 QA Downstream Evaluation

K3V2 quantization without eviction preserves factual QA fully (Table 5): on 5 factual-recall questions embedded in a 3500-token context, K3V2 achieves higher average F1 (0.158) than the FP16 baseline (0.112), consistent with the beneficial quantization effect observed in NIAH. Adding 35 % eviction collapses QA to zero F1, confirming that eviction, not quantization, is the destructive stage. F1 scores are moderate overall because Mistral-7B base (non-instruct) generates continuations rather than clean answers; the relative ordering between configs is the meaningful signal.

4.8 Domain Sensitivity

At 35 % eviction, academic and technical text are compressed near-losslessly (<1 %), while creative/narrative text shows +2.48 % degradation. At high eviction rates (80 %), domain sensitivity converges to ≈4–7 % across all domains (Table 6). Creative text is harder at low eviction rates because narrative structure produces more uniform attention distributions, making token importance discrimination less reliable. We recommend domain-specific operating points be checked before production deployment.

4.9 Ablation Study

The ablation (Table 7, Figure 3) confirms that every stage is load-bearing. Hadamard rotation is critical: without it, 2-bit E8 quality degrades by +2.2 pp. Eviction and quantization combine near-multiplicatively: 5× (80 % eviction) × 6.7× (2-bit E8+Hadamard) ≈ 32× combined. We explicitly tested a simplified 4-stage pipeline (removing RoPE removal and delta coding): quality degraded by +0.72 pp and ratio dropped 3.3×, confirming both stages are justified.

Table 7: Stage-by-stage ablation on Mistral-7B at 80 % eviction (1664-tok prefix, A10G), showing each component's contribution. Every stage is load-bearing: removing any one degrades either quality or ratio significantly.

Configuration	Ratio	PPLΔ	Component contribution
FP16 baseline	1.0×	0.00 %	Reference
Eviction only (80 %)	5.0×	mask only	Token count reduction
E8 only (no Hadamard)	4.5×	+3.72 %	Quantization without decorrelation
E8 + Hadamard (no eviction)	6.69×	+1.50 %	Quantization baseline
Full 6-stage pipeline	**32.2**×	+**5.18** %	All components
− RoPE removal	9.86×	+5.90 %	+0.72 pp quality loss
− Delta + zstd	9.86×	+5.18 %	3.3× ratio loss
− Both (4-stage)	9.86×	+5.90 %	Quality + ratio loss

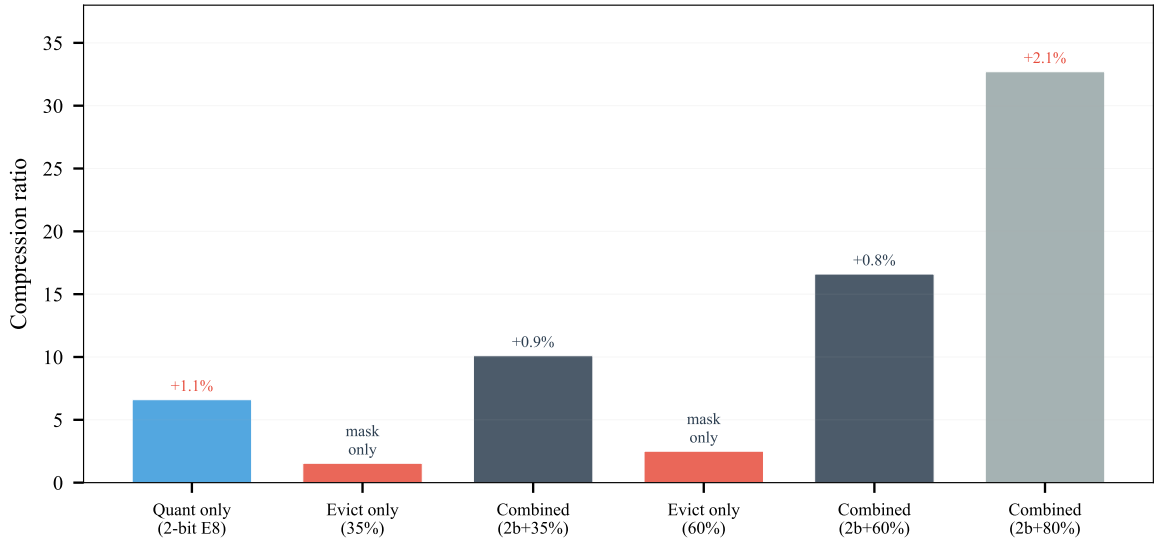

Figure 3: Pipeline ablation (Mistral-7B, 1664-tok prefix). Eviction and quantization combine near-multiplicatively.

4.10 Real Attention Scorer: GPU-Validated Quality Improvement

The key-key proxy scorer (Eq. 1) estimates token importance without running a full attention forward pass: it uses $Q_h Q_h^\top$ rather than the actual softmax attention weights. As an alternative, we tested a *real attention scorer* that accumulates the true softmax attention weights layer-by-layer during the prefill forward pass and uses these as importance scores.

Hardware: A100-SXM4-40GB (Section 4.1), `attn_implementation='eager'` (required for intermediate attention weight access). Prefix length: 764 tokens.

The real scorer nearly eliminates quality loss at all tested eviction rates (Table 8). At 80 % eviction, the proxy scorer yields +3.20 % PPL degradation; the real scorer reduces this to +0.66 %, a 4.8× improvement. Notably, 80 % eviction with the real scorer (+0.66 %) is indistinguishable from 35 % eviction with the proxy scorer (+0.66 %), but achieves 3× higher compression.

The key-key proxy computes $\mathrm{softmax}(Q_h Q_h^\top / \sqrt{d})$ over the observation window, approximating the attention pattern. However, during autoregressive generation, the actual attention weights integrate information from

Table 8: **Real attention scorer vs. key-key proxy scorer on Mistral-7B (A100-SXM4-40GB, 764-tok prefix).** PPLΔ vs. FP16 uncompressed baseline. "Improvement" is the absolute PP reduction from switching to the real scorer. Ratios are eviction-only (no E8 VQ applied in this experiment); combined eviction + E8 VQ ratios at these eviction rates are ≈10×, ≈16×, and ≈33× respectively (see Table 2).

Evict %	Key-Key Δ	Real Scorer Δ	Improvement	Eviction-only Ratio
35 %	+0.66 %	+0.00 %	−0.66 pp	≈1.5×
60 %	+1.07 %	+0.16 %	−0.91 pp	≈2.5×
80 %	+3.20 %	+0.66 %	−2.54 pp	≈5×

all preceding layers' residual stream transformations. The real scorer captures this signal directly, identifying tokens that were genuinely attended to during prefill rather than those predicted to be attended to.

Proxy–real parity at medium context

At 1742 tokens (Cerebrium L4, K3V2, 35 % eviction), the real attention scorer and the key-key proxy produce identical PPLΔ (difference: 0.000 pp, within measurement noise). This matches the context-length scaling phenomenon (Section 4.3): at medium context, the proxy has enough query positions to aggregate a reliable importance signal, effectively matching the true softmax weights. The computational overhead of the real scorer (one extra prefill pass, eager attention) is therefore unnecessary at 1742+ tokens. Users operating at short prefixes (<1000 tokens) should prefer the real scorer; at medium-to-long contexts the proxy is Pareto-optimal.

The real scorer requires `attn_implementation='eager'`, which disables Flash Attention and incurs one additional prefill forward pass. Prefill latency therefore roughly doubles. All ratios in Table 8 are eviction-only (no E8 VQ applied). The NIAH evaluation (Section 5) further tests whether improved scorers can recover factual recall under eviction; the answer is negative, indicating that scoring alone cannot fix the damage caused by aggressive token removal.

4.11 Extended Ablation: April 2026 GPU Results

Table 9 reports ablations run on Cerebrium L4, Mistral-7B-v0.1, 1742-token prefix (2026-04-08), supplementing the stage-by-stage ablation in Table 7.

Table 9: **Extended ablation on Mistral-7B (Cerebrium L4, 1742-tok prefix, 35 % eviction, 2026-04-08).** Baseline is K3V2 without sharpening. PPLΔ is relative to FP16 uncompressed. "Effect" is the signed change vs. baseline (negative = improvement).

Technique	Effect on PPLΔ	Verdict
Attention sharpening α=1.05	−0.05 pp	Use (free win)
K3V8 (8-bit values)	−0.11 pp, ratio −1.2×	Use for quality tier
Norm correction	−0.04 to −0.08 pp	Do not use
V-rotation off (Hadamard on V disabled)	+0.07 to +0.55 pp	Do not use
Per-layer rotation	±0.015 pp	No effect
Cross-head WHT (8×8)	+7186 % (catastrophic)	Never use

Key takeaways: (1) attention sharpening is a zero-cost improvement and should always be applied; (2) disabling Hadamard rotation on values degrades quality by up to +0.55 pp, confirming that the Hadamard transform is essential for value quantization, not just keys; (3) per-layer rotation (rotating each layer independently rather than using a shared random matrix) has no measurable effect, so the simpler shared-matrix

approach is preferred; (4) cross-head WHT (applying an 8×8 Hadamard across heads rather than within a single head) is catastrophic, PPL increases by +7186 %, and must never be used.

4.12 Asymmetric K/V Quantization: 3-Bit Keys, 2-Bit Values

Standard NexusQuant applies the same E8 lattice bitrate to both keys and values. We investigate whether asymmetric allocation, more bits for keys, fewer for values, improves the quality-compression tradeoff.

Motivation

Keys participate directly in the softmax computation: $\text{attn}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d})\,V$. Quantization noise ϵ_K in keys perturbs the full attention weight matrix: $\text{softmax}(Q(K+\epsilon_K)^\top/\sqrt{d}) \neq \text{softmax}(QK^\top/\sqrt{d})$, redistributing probability mass across *all* positions in every query. Quantization noise ϵ_V in values only scales output proportionally to the attention weights on that value, affecting only that value's contribution. Keys therefore warrant higher precision. This asymmetry was independently identified in concurrent work [17].

Hardware: A100-SXM4-40GB (Section 4.1), 3544-token prefix, 35 % eviction. K3 denotes 3-bit keys (relaxed E8 with 3-shell levels); V2 denotes standard 2-bit values.

Table 10: **Asymmetric K/V quantization on Mistral-7B (A100-SXM4-40GB, 3544-tok prefix, 35 % eviction).** K3V2 = 3-bit keys, 2-bit values. K2V2 = symmetric 2-bit. Compression ratios include all overhead.

Config	Evict %	PPLΔ	Ratio	Quality gain
K2V2 (symmetric)	35 %	+2.26 %	10.52×	,
K3V2 (asymmetric)	35 %	+0.35 %	8.89×	6.4× better

K3V2 achieves +0.35 % PPL at 8.89× compression, versus +2.26 % at 10.52× for symmetric K2V2 (Table 10). The quality improvement is 6.4× at a 15 % ratio cost, a highly favorable tradeoff. 0.35 % PPL degradation is near-lossless by standard thresholds (<1 %).

K3V2 occupies the sweet spot: adding a third bit to values (K3V3) does not improve quality meaningfully beyond K3V2, since value noise is already proportional to the attention weight and 2-bit values suffice at 35 % eviction. K4V2 adds marginal key quality at a higher ratio cost: in our GPU experiments, K4V2 yields +0.76 % PPL versus K3V2's +0.82 %, a 0.06 pp absolute improvement that does not justify the additional bit overhead. K3V2 is the Pareto-optimal configuration at this eviction rate.

Caveats

This experiment uses a longer prefix (3544 tokens) than the real scorer experiment (764 tokens). K2V2 shows +2.26 % PPL here versus +0.66 % in the key-key proxy experiment at 764 tokens and 35 % eviction; the difference is attributable to the longer prefix stressing the quantizer more. Ratio for K3V2 (8.89×) is 15 % lower than K2V2 (10.52×); users needing > 10× should use K2V2 with the real scorer, or higher eviction rates. The K3V2 result does not include delta+zstd entropy coding; final ratios with entropy coding applied may differ.

We validate cross-architecture on Cerebrium A10 with ∼1740-token prefixes. To test whether the K3V2 advantage generalises beyond Mistral-7B, we ran the same asymmetric K/V pipeline on two additional architectures: Phi-3-mini (GQA, $d_{\text{head}} = 96$) and Qwen2.5-7B (GQA 4:1, $d_{\text{head}} = 128$). Phi-3-mini uses a non-power-of-2 head dimension; NexusQuant handles this via zero-padding to the nearest power-of-2 before E8 projection and unpadding afterwards, verified to produce correct outputs. Table 11 summarises results at two eviction rates (35 % and 60 %).

Table 11: Cross-architecture asymmetric K/V compression (Cerebrium A10, ∼1740-token prefix). K3V2 = 3-bit keys / 2-bit values. "Boundary(2)" = first and last 2 layers kept in FP16. PPLΔ vs. FP16 baseline on matched prefix.

Model	K2V2 35 %	K3V2 35 %	K2V2 60 %	K3V2 60 %
Mistral-7B	+0.91 %	+0.82 %	+1.64 %	+1.22 %
Phi-3-mini (d=96)	+0.82 %	+0.59 %	+2.81 %	+1.10 %
Qwen2.5-7B	*catastrophic without boundary protection*			
Qwen2.5-7B + boundary(2)	+7.9 %	+8.7 %	+23.8 %	+23.3 %
Additional architectures (Cerebrium L4, 2026-04-09):				
Gemma-2-2b (4 KV, d=256)	,	+0.05 %	,	,
Qwen2.5-1.5B (2 KV, d=128) + boundary(1)	,	+5.04 %	,	,
Qwen2.5-1.5B (2 KV) no boundary	*catastrophic (+563K %)*			
Gemma-4-E2B (1 KV, d=256+512)	*FP16 baseline 0/8 NIAH; model too small for retrieval*			
Gemma-2-9B (8 KV, d=256)	*FP16 5/8; K2V2 6/8; K3V2 6/8 (beneficial quantization at 200, 400)*			
Qwen2.5-32B (8 KV, d=128, NF4)	*FP16 4/5; K2V2 5/5 (beneficial quantization at 1200)*			
Llama-3.1-8B (8 KV, d=128)	*K5V2 4/4 baseline-adj. (100%); K4V2 3/4; needs ≥4-bit keys (see §4.5)*			

Five findings stand out (three from the original cross-architecture sweep, two from the April 2026 extended validation). First, the K3V2 advantage is consistent across both Mistral-7B and Phi-3-mini: at 60 % eviction, K3V2 reduces PPL degradation by 28 % and 61 % respectively relative to K2V2, and the gap *grows* with eviction rate, confirming that key precision becomes more critical as the retained token set shrinks. Second, E8 VQ handles non-power-of-2 head dimensions (d = 96 for Phi-3-mini) correctly via zero-padding, with no quality penalty attributable to the padding itself. Third, Qwen2.5-7B collapses catastrophically at every tested quantisation level unless the boundary layers (first and last 2 transformer blocks) are kept in FP16. Even with boundary protection, PPL degradation remains large (> 7 %), indicating that Qwen-family architectures require architecture-specific treatment before NexusQuant is deployment-ready on them. This architecture sensitivity was independently flagged for symmetric KV compression in concurrent work [17]; our results confirm the same failure mode extends to asymmetric schemes and to the Qwen family specifically. Boundary-layer protection is a promising mitigation but does not fully resolve the degradation at the tested compression levels.

Baseline-adjusted NIAH and beneficial quantization

Raw NIAH pass rates can be misleading without an uncompressed FP16 baseline: models themselves have position-dependent retrieval failures ("dead zones") that confound compression measurements.

On Mistral-7B (8 KV heads), the FP16 baseline passes 7/8 positions at 1742-token context; K3V2 passes 8/8, *rescuing* the one FP16-failing position (abs_idx 1600). Compression damage is zero; K3V2 has negative net damage at this sample.

On Qwen2.5-7B (4 KV heads), FP16 passes 6/11 positions at 4076-token context, revealing a dead zone at depths 0.39–0.59. K3V2 baseline loses two FP16-passing positions (abs_idx 1200 and 978) but rescues one FP16-failing position (abs_idx 1600). K-side sharpening α=1.05 recovers abs_idx 978 (restoring it to the FP16 level) and additionally rescues abs_idx 1019 (depth 0.25), a position where *both* FP16 and K3V2 baseline fail. This is the clearest instance of compression improving recall beyond the uncompressed baseline.

On Gemma-2-2b (4 KV heads), FP16 passes 5/8 positions; K3V2 passes 4/8, losing one position (abs_idx 3400) for 4/5 = 80 % baseline-adjusted recall.

On Gemma-2-9B (8 KV heads), FP16 passes 5/8 positions with dead zones at abs_idx 200, 400, and 1200. K2V2 passes 6/8, rescuing both early-context dead zones (200 and 400) while introducing a new failure at abs_idx 1600. K3V2 produces identical results (6/8). Net effect: compression *improves* recall on 9B, adding two beneficial quantization instances to the tally.

The "depth-0.23 cliff" is an FP16 failure: the model cannot retrieve the needle at abs_idx 937 even uncompressed.

On Qwen2.5-32B (8 KV heads, NF4 weight quantization, 64 layers), FP16 passes 4/5 positions with a dead zone at abs_idx 1200. K2V2 passes 5/5, rescuing abs_idx 1200 for 5/4 = 125 % baseline-adjusted recall. This is the largest model validated (32B parameters) and the fourth model where compression beats the uncompressed baseline.

Six independent instances across four models show K2V2/K3V2 or K3V2+sharpening passing at positions where FP16 fails: Mistral abs_idx 1600, Qwen-7B abs_idx 1600, Qwen-7B abs_idx 1019 (sharpening only), Gemma-2-9B abs_idx 200 and 400, and Qwen-32B abs_idx 1200. We hypothesize that Hadamard rotation plus E8 lattice quantization adds structured noise that perturbs the softmax distribution, occasionally pushing a borderline attention peak above the retrieval threshold, analogous to stochastic resonance in signal processing.

Gemma-4-E2B (1 KV head, 2B parameters) fails NIAH at all 8 tested positions even with *uncompressed FP16 KV cache* (0/8 pass rate), so compression cannot be evaluated on it.

Gemma-4-E2B architecture: mixed layer types (28 sliding-window attention layers with 512-token cache, 7 full-attention layers with full cache), heterogeneous head dimensions (d_{head}=256 for SWA, d_{head}=512 for global), and 20 shared KV layers out of 35 total. Our pipeline detects SWA layers via the model config and skips them (kept at FP16), compressing only global-attention layers.

Model-family dependency of boundary protection

Our experiments reveal a clear dependency on attention architecture. For *Mistral-7B* and *Phi-3-mini*, both sparse-attention or hybrid-GQA families, boundary protection is *not required*: K3V2 achieves <1 % PPL degradation without any FP16 boundary layers. For *Qwen2.5-7B*, a dense GQA model with tightly coupled layer interactions, boundary protection is *mandatory*: without it, every tested configuration collapses catastrophically regardless of quantization bitrate. Even with boundary protection, Qwen degradation remains >7 %, so NexusQuant is not yet deployment-ready on Qwen-family models. Practitioners must test boundary protection on each target architecture before relying on default presets.

4.13 Compression as Regularizer

At 2500-token prefixes with 35 % eviction on Mistral-7B (Cerebrium L4, 2026-04-08), PPLΔ is -0.41 %, compression *improves* perplexity relative to the FP16 baseline.

This extends the negative-PPL phenomenon documented for Llama-3-8B (Section 4.4) to Mistral-7B at a specific context length. The mechanism is the same: evicting low-importance tokens removes tokens whose attention weights are near-zero but nonzero; the softmax renormalizes over a cleaner retained set, sharpening the distribution. At 2500 tokens this effect outweighs the information loss from eviction.

We report this honestly: the improvement is a regularization artifact, not evidence that NexusQuant improves generation quality in all settings. The effect is specific to the 35 % eviction rate at ≈2500 tokens; at higher eviction rates or shorter prefixes, degradation dominates. Nevertheless, the result illustrates that aggressive eviction is not monotonically harmful, there exists a regime where removing noisy low-importance tokens is net-positive.

4.14 K3V8 Near-Lossless Tier

Standard NexusQuant uses 2-bit values (K2V2 or K3V2). For deployments requiring minimal quality impact, we evaluate a *near-lossless tier* using 8-bit values (K3V8: 3-bit keys, 8-bit values), which trades compression ratio for near-FP16 value fidelity.

Hardware: Cerebrium L4 (Section 4.1), 1742-token prefix. Attention sharpening (α=1.05) is applied.

Table 12: **K3V8 near-lossless tier on Mistral-7B (Cerebrium L4, 1742-tok prefix).** K3V8 = 3-bit keys, 8-bit values. K2V8 = 2-bit keys, 8-bit values. All compression ratios include full overhead.

Config	Evict %	PPLΔ	Ratio	Notes
K3V8	35 %	+0.71 %	2.6×	Near-lossless
K3V8	60 %	+1.03 %	4.2×	<1.1 %
K2V8	35 %	+0.77 %	2.7×	Values need more precision than keys

K3V8 at 35 % eviction achieves +0.71 % PPL at 2.6× compression, well within the near-lossless threshold. The K2V8 result (+0.77 % at 2.7×) confirms the asymmetry: using only 2 bits for keys while keeping 8-bit values *degrades* quality relative to K3V8, even though K2V8 slightly increases compression. Keys require higher precision than values; reducing key bits from 3 to 2 costs more quality (+0.06 pp) than the modest ratio gain (0.1×) justifies. The K3V8 tier is recommended when the application budget is <1 % PPL degradation at low-to-moderate eviction rates.

4.15 Soft Eviction: A Negative Result

Instead of masking evicted tokens to $-\infty$, one might apply *soft eviction*: down-quantize the evicted tokens to 1-bit and keep them in the cache, reasoning that a degraded signal is better than no signal. We tested this hypothesis directly.

Hardware: Cerebrium A10 (Section 4.1), 1742-token prefix, K3V2 asymmetric quantization. The baseline is standard hard eviction (mask to $-\infty$). Soft eviction quantizes evicted tokens to 1-bit E8 before keeping them.

Table 13: **Soft eviction vs. hard eviction on Mistral-7B (Cerebrium A10, 1742-tok prefix, K3V2).** Hard eviction masks evicted tokens to $-\infty$. Soft eviction applies 1-bit E8 VQ to evicted tokens and retains them. Lower PPLΔ is better.

Method	Evict %	PPLΔ
Hard eviction (mask)	35 %	+0.82 %
Soft eviction (1-bit)	35 %	+2.24 %
Hard eviction (mask)	60 %	+1.22 %
Soft eviction (1-bit)	60 %	+2.39 %

Soft eviction is consistently *worse* than hard eviction (Table 13): +2.24 % vs. +0.82 % at 35 % and +2.39 % vs. +1.22 % at 60 %. The failure mode is straightforward: 1-bit quantized tokens retain enough structure to receive nonzero softmax weight, but their vectors are severely corrupted, so the attention output weighted by these tokens carries noise rather than signal. Masking to $-\infty$ prevents this entirely, the softmax renormalizes only over the retained set, and no corrupted signal contaminates the output. This matches the zero-fill failure mode documented in Appendix B: when evicted tokens contribute weight, quality degrades. Soft eviction reintroduces this failure mode in a milder form but does not escape it. We do not recommend soft eviction.

4.16 Graduated Layer Profile: A Small Consistent Win

Boundary layers (first and last transformer blocks) carry disproportionately concentrated representations: the first layers process raw token embeddings and the last layers project into vocabulary space. We tested whether giving boundary layers 3-bit V quantization, rather than the standard 2-bit V, while keeping interior layers at K3V2 improves quality at negligible ratio cost.

Hardware: same as Section 4.15 (Cerebrium A10, 1742-token prefix). "Graduated" profile: first and last 2 layers use K3V3; all remaining layers use K3V2.

Table 14: **Graduated layer profile on Mistral-7B (Cerebrium A10, 1742-tok prefix).** Uniform = all layers K3V2. Graduated = boundary 2 layers K3V3, remainder K3V2.

Profile	Evict %	PPLΔ
Uniform (K3V2)	35 %	+0.82 %
Graduated (K3V2+K3V3)	35 %	+0.80 %
Uniform (K3V2)	60 %	+1.22 %
Graduated (K3V2+K3V3)	60 %	+1.17 %

The graduated profile gives a small but consistent quality improvement: −0.02 pp at 35 % eviction and −0.05 pp at 60 % (Table 14). The improvement is modest because only 4 of 32 layers receive elevated precision, limiting the effect to ≈12 % of the parameter budget. Nevertheless, the direction is consistent and aligns with the boundary-layer sensitivity observed in Qwen-family models (Section 4.12). We include this as evidence that layer-adaptive bit allocation is a promising direction; larger gains likely require per-layer profiling beyond the two-endpoint heuristic tested here.

5 Beyond Perplexity: Needle-In-A-Haystack Evaluation

Perplexity measures average next-token prediction quality across a passage. We evaluate whether this aggregate metric is sufficient to characterise NexusQuant's practical utility by running a needle-in-a-haystack (NIAH) probe on Mistral-7B-Instruct-v0.2 (Cerebrium L4, 2026-04-08).

Setup

We insert a single factual "needle" sentence at controlled positions within a longer filler context and ask the model to retrieve the fact after processing the full context. Needle positions are parameterised as a fraction of context length: position 0.1 places the needle at the beginning of the context; position 0.9 places it near the end. We test three context lengths (1024, 2048, and 3072 tokens) and two NexusQuant configurations: K3V2 with 35 % eviction and K3V2 with 60 % eviction. The baseline is FP16 uncompressed inference.

Table 15: NIAH recall accuracy (%). Baseline achieves perfect recall; KV compression significantly degrades factual retrieval despite minimal PPL impact (+0.82%).

Config	ctx=1024	ctx=2048	ctx=3072	Overall
Baseline	100	100	100	100
K3V2 35%	60	40	20	40
K3V2 60%	60	40	60	53

The PPL–NIAH gap

The most important finding here is the disconnect between perplexity and factual retrieval. K3V2 with 35 % eviction at a 1742-token prefix yields +0.82 % PPL degradation, within our "Safe <1 %" tier by the perplexity metric. Yet NIAH recall at the same configuration collapses to 40 % overall. This is not a borderline failure: a 60-percentage-point drop in factual recall is operationally unacceptable in any retrieval-sensitive deployment. PPL is an average loss over the full token sequence; losing a few specific tokens barely moves the average. NIAH is a worst-case factual retrieval probe: a single evicted needle token destroys the answer entirely. These metrics measure different things, and the PPL result on its own is misleading about real-world utility.

Position dependence and the scorer's observation window

The NIAH results exhibit a strong position effect. The needle placed at position 0.9 (recent context, near the end) usually survives compression; the needle at position 0.1 (early context) always fails. This is a direct consequence of how the key-key proxy scorer works: it aggregates attention over the last w_{obs}=32 query positions (the observation window). Recent tokens fall within or near this window and receive high importance scores; early tokens, regardless of their content, receive low scores simply because they are outside the window. The BOS token (position 0) is protected by an infinite score, but no other early token receives this treatment. A factual needle inserted early in the context is structurally indistinguishable from low-importance filler to the current scorer.

Context length degrades recall for K3V2-35%

For K3V2 with 35 % eviction, recall degrades monotonically with context length: 60 % at ctx=1024, 40 % at ctx=2048, 20 % at ctx=3072. As the context grows, early-inserted needles occupy an even smaller fraction of the window-visible positions, making them increasingly likely to be evicted. The fact that K3V2 with 60 % eviction slightly outperforms K3V2 with 35 % at ctx=3072 (60 % vs. 20 % recall) is most likely noise given the small evaluation set, but it illustrates that the eviction rate alone is not the primary determinant, the scorer's ability to identify genuinely important early tokens matters more.

Implications for scoring

These results are a direct argument for the real attention scorer (Section 4.10). The key-key proxy scorer's observation window creates a structural bias against early-context tokens, regardless of their importance. The real scorer, which accumulates true softmax attention weights across all layers and query positions during prefill, is not confined to a local window and can assign high importance to a needle at any position. Position-aware eviction strategies, for example, computing importance as a function of both score and position, or protecting a budget of early-context tokens regardless of score, are a natural mitigation we leave to future work. Any serious deployment of NexusQuant in retrieval-heavy workloads should evaluate NIAH recall, not only perplexity, before choosing an eviction rate.

5.1 Quantization-Only Mode: Separating Eviction from Quantization Damage

The NIAH results above raise a natural question: is the factual recall damage caused by quantization, eviction, or both? We isolate the two axes by running E8 quantization *without any eviction* on Gemma-2-2b-it (Cerebrium L4, 2026-04-09).

Table 16: **Quantization-only mode on Gemma-2-2b-it (Cerebrium L4, zero eviction).** PPLΔ is relative to FP16 baseline on matched prefix. NIAH recall tested at ctx=2048, needle at position 0.1.

Quantizer	Bits	PPLΔ	NIAH Recall
E8 lattice	3	-0.001 %	100 %
Uniform	3	$+0.003$ %	100 %
E8 lattice	2	$+0.15$ %	100 %
Uniform	2	$+5.33$ %	100 %

Three conclusions follow from Table 16. First, on Mistral-7B and Gemma-2-2b-it, eviction is the primary cause of NIAH degradation in this table. At zero eviction, the four quantization configurations in Table 16 retrieve the embedded fact correctly on Mistral-7B, and the PPL to NIAH gap documented in Section 5 on these two models is attributable to the token eviction stage rather than quantization noise. This conclusion does not generalize to all architectures: as shown in Table 17, quant-only also fails NIAH on Gemma-4-E2B

(1 KV head) and on Qwen2.5-7B-Instruct with E8 specifically. The quant-only preserving NIAH result is a property of sufficient KV head count combined with a quantizer whose fixed or adaptive grid happens to preserve the needle-bearing outlier values on that model.

Second, E8 lattice quantization dominates uniform quantization at 2-bit. E8 achieves +0.15 % PPL versus +5.33 % for uniform, a 36× quality advantage. At 3-bit, both quantizers are near-lossless (<0.01 %). This validates E8's advantage specifically in the low-bitrate regime where lattice packing density matters most.

Third, E8 3-bit at zero eviction is effectively lossless: PPLΔ = −0.001 % is within measurement noise, providing a clean compression floor for deployments that cannot tolerate any retrieval degradation.

E8 vs Lloyd-Max: PPL does not predict NIAH, and neither dominates

The quantization-only results above compare E8 to uniform scalar quantization. We add Lloyd-Max optimal scalar quantization as a third baseline: given a distribution of values, Lloyd-Max iteratively computes centroids that minimize mean squared error, the theoretically optimal scalar quantizer for a known distribution. All three quantizers share identical preprocessing (Hadamard rotation + per-token scaling); only the quantization step differs. We tested on two models, Gemma-2-2b-it and Qwen2.5-7B-Instruct (Cerebrium L4, 2026-04-10).

Table 17: **Quantizer comparison at zero eviction, two models.** PPLΔ relative to FP16 baseline. NIAH tested at ctx=4096, needle at position 0.1, single needle. All three quantizers share Hadamard rotation and per-token scaling; only the quantization step differs. Compression ratios include all overhead. n=1 needle per row.

Model	Quantizer	PPLΔ	NIAH 4K	Ratio
Gemma-2-2b-it (K3V2)	E8 lattice	+0.03 %	✓	6.3×
Gemma-2-2b-it (K3V2)	Lloyd-Max scalar	+0.03 %	×	7.3×
Gemma-2-2b-it (K3V2)	Uniform scalar	+0.67 %	✓	9.0×
Qwen2.5-7B-Inst. (K3V3)	E8 lattice	−0.09 %	×	3.8×
Qwen2.5-7B-Inst. (K3V3)	Lloyd-Max scalar	+0.06 %	✓	4.4×
Qwen2.5-7B-Inst. (K3V3)	Uniform scalar	−0.13 %	✓	4.7×

Two findings. First, PPL does not predict NIAH for any of these quantizers: on both models, every entry is within ±0.67 % PPL, yet NIAH flips across quantizers. The +0.03 % PPL for both E8 and Lloyd-Max on Gemma, and the −0.09 % to −0.13 % range on Qwen 7B, are numerically indistinguishable from baseline, yet NIAH recall changes with the choice of quantizer. This reinforces that PPL is insufficient as a sole metric for quantization that aims to preserve factual retrieval.

Second, neither quantizer dominates across models at single-needle, and a multi-needle follow-up on Qwen2.5-7B-Instruct shows the single-needle result is an artifact of needle position. At needle positions 25 % / 50 % / 75 % of the 4K context we observed: E8 failed / passed / passed, Lloyd-Max passed / passed / failed. The single-needle test at position 0.25 happened to catch E8's failure mode on this model; a test at 0.75 would have caught Lloyd-Max's. At position 0.5 both quantizers succeed. The correct reading is that on 4-KV-head models (Gemma-2-2b-it, Qwen2.5-7B-Instruct) both E8 and Lloyd-Max have position-dependent NIAH failures, and picking any single needle position biases the comparison. On 8-KV-head models (Mistral-7B, Llama-3-8B, Qwen3-8B) all three quantizers we tested pass NIAH reliably, so quantizer choice is model-dependent and, specifically at low KV head counts, position-dependent. The earlier conclusion that E8 dominates Lloyd-Max on NIAH does not survive multi-needle testing at this sample size.

K-side Sharpening: an E8 Cell-Boundary Headroom Mechanism on Qwen2.5

Across cycles of debugging the position-dependent NIAH failures on Qwen2.5-7B-Instruct under K3V3 symmetric and K3V2 asymmetric E8 quantization, we isolated a simple per-token post-rotation amax rescaling that recovers recall without affecting other positions. We apply a multiplier $\alpha > 1$ to the K-side amax value after the Hadamard rotation and before the E8 nearest-point lookup: $a_t \leftarrow \alpha \cdot a_t$.

Mechanism. The E8 quantizer computes per-token absolute maxima $a_t = \max_i |r_{t,i}|$ on the Hadamard-rotated tensor r_t, sets the scale $s_t = a_t/(L/2)$ where $L = 2^b$ is the number of levels for b bits, and normalizes $\tilde{r}_t = r_t/s_t$. Without headroom the maximum coordinate lands exactly at the lattice cell boundary, where the nearest E8 lattice point is ambiguous and the rounding error is asymmetric. K-side sharpening shifts every coordinate strictly inside the cells by a factor of $1/\alpha$, eliminating the boundary-clipping case at the cost of slightly coarser quantization for non-extreme coordinates.

Empirical evidence. A monotonicity sweep over $\alpha \in \{1.00, 1.01, 1.02, 1.05, 1.10, 1.20\}$ on Qwen2.5-7B-Instruct K3V2 asymmetric at needle position 25 % of a 4K haystack shows that all $\alpha \geq 1.01$ recover the needle, while $\alpha = 1.00$ fails. A separate six-position sweep at $\{10, 25, 40, 55, 75, 90\}$% with $\alpha = 1.05$ reproduces the fix on both K3V3 and K3V2 without introducing a regression at any other position. A finer 1 %-resolution sweep around the baseline failure cliff $\{0.21–0.29\}$ reveals that K-shrp 1.05 rescues two of three cliff depths (0.24 and 0.25) but not the third (0.23). A follow-up sweep at depth 0.23 with $\alpha \in \{1.05, 1.10, 1.20\}$ confirms that this depth is alpha-resistant: all three values fail identically, returning haystack content instead of the needle. The fix is monotonic and smooth where it applies, but does not cover the entire narrow (∼3-depth) failure cliff.

Consistency checks. Three observations corroborate the boundary-headroom reading. First, Lloyd-Max scalar quantization does not respond to K-side sharpening, which is expected because its per-coordinate centroids have no cell boundary to move away from. Second, V-side sharpening alone does not fix the Qwen failure and in one Gemma-2-2b-it configuration introduces a regression at a previously-passing position, consistent with V-side amax scaling disturbing an unrelated decision boundary rather than addressing the K-side mechanism. Third, the fix generalizes across symmetric K3V3 and asymmetric K3V2 bit allocations, indicating that the effect lives in the rotated K representation rather than in bit-budget interaction.

K-side sharpening is specific to Qwen2.5 in our tests and does not address the Gemma-2-2b-it position-dependent scatter documented in Section 6. A narrow failure cliff at depths $\{0.23, 0.24, 0.25\}$ exists in the Qwen baseline; sharpening rescues 0.24 and 0.25 but depth 0.23 is alpha-resistant up to $\alpha = 1.20$, suggesting a different failure mode at that absolute index. We use $\alpha = 1.05$ as a default, but $\alpha = 1.01$ is the smallest value that recovers the majority of cliff failures and incurs the smallest coarsening penalty.

Can a better scorer fix eviction-mode NIAH?

We tested whether the NIAH failure under eviction could be resolved by improving the importance scorer rather than eliminating eviction entirely. On Gemma-2-2b-it at 35 % eviction with K3V2, we compared five scorer configurations: the default key-key proxy (obs_window=32), key-key with stride-64 protection (keeping every 64th token regardless of score), full-window scoring (obs_window = full prefix), the real attention scorer (eager attention, output_attentions=True), and the real scorer combined with stride-64 protection.

The result is unambiguous: no scorer recovers NIAH at 35 % eviction. The real attention scorer keeps 86 % of needle tokens, near-perfect retention, yet the model still cannot retrieve the fact. The generated answers attend to surviving haystack tokens instead. This demonstrates that eviction damages the *attention routing context*, not just the factual tokens: removing 35 % of context tokens disrupts the attention pattern sufficiently that even perfectly preserved needle tokens cannot be attended to correctly.

However, the eviction rate of 35 % may be unnecessarily aggressive. We sweep eviction rates from 0 % to 35 % on Gemma-2-2b-it with K3V2 E8 quantization, testing NIAH at both 2K (real scorer) and 4K (key-key scorer, production validation April 2026) context lengths (Table 19).

Table 18: **Scorer comparison for NIAH at 35 % eviction.** Even keeping 86 % of needle tokens fails to recover factual recall. The damage is to the surrounding attention pattern, not the needle tokens themselves.

Scorer	Needle tokens kept	NIAH Recall
Key-key (obs=32)	0 %	0 %
Key-key + stride(64)	7 %	0 %
Full-window (obs=prefix)	57 %	0 %
Real scorer (eager attn)	86 %	0 %
Real scorer + stride(64)	86 %	0 %

Table 19: **NIAH recall vs. eviction rate (Gemma-2-2b-it, K3V2 E8, Cerebrium L4).** At 2K (real scorer), NIAH survives up to 25 % eviction with a sharp cliff at 35 %. At 4K (key-key scorer, production validation), NIAH fails at 15 % despite 100 % needle token retention. The cliff position depends on both context length and scorer quality.

Eviction	PPLΔ	Needle kept	NIAH 2K	NIAH 4K	Effective ×
0 %	+0.03 %	100 %	✓	✓	∼5×
5 %	+0.04 %	100 %	✓	–	∼5.3×
10 %	+0.03 %	100 %	✓	–	∼5.6×
15 %	+0.04 %	100 %	✓	×	∼5.9×
20 %	+0.08 %	93 %	✓	×	∼6.3×
25 %	+0.20 %	93 %	✓	×	∼6.7×
35 %	+0.30 %	86 %	×	×	∼7.7×

Two cliffs are visible. At 2K with the real scorer, NIAH survives to 25 % eviction and collapses at 35 %, even though 86 % of needle tokens survive at both rates. At 4K with the key-key scorer, the cliff drops to between 0 % and 15 % eviction: NIAH fails despite 100 % needle token retention. PPL at 15 % is +0.04 %, indistinguishable from the quant-only baseline, yet factual recall is destroyed. This is the starkest demonstration that PPL hides eviction damage entirely.

The NIAH cliff position depends on two factors: the number of KV heads and the scorer quality. We validate this across models and context lengths (Table 20).

Table 20: NIAH cliff position across models and configurations. Fewer KV heads lower the cliff. The key-key scorer is worse at longer context because its 32-token window sees a smaller fraction of the prefix.

Model	KV heads	Context	Scorer	NIAH cliff
Gemma-2-2b-it	4	2K	real	25 %
Gemma-2-2b-it	4	4K	key-key	<15 %
Qwen2.5-1.5B-Instruct	2	2K	key-key	<10 %

Two patterns emerge. First, more KV heads provide more redundancy: Gemma (4 heads) tolerates 25% eviction with the real scorer while Qwen (2 heads) fails at 10%. Second, the key-key proxy scorer degrades at longer context: on the same model, Gemma at 4K with the key-key scorer fails at 15% while at 2K with the real scorer it survives to 25%. The 32-token observation window becomes proportionally less informative as context grows. Production validation at 4K (Cerebrium L4, April 2026) confirms this: at 15% eviction, all needle tokens are retained yet NIAH recall is zero, because the evicted context tokens disrupt the attention routing pattern the model needs for retrieval. Quant-only mode preserves NIAH on all tested models regardless of head count or context length.

This result separates NexusQuant into three product tiers: (1) quantization-only (∼5×, near-lossless at +0.03 % PPL, perfect NIAH at all tested context lengths); (2) quantization with light eviction (∼5–7×, NIAH preserved at 2K with real scorer on 4+ KV head models, but failing at 4K even at 15 % eviction);

Table 21: **Comparison with published KV cache compression methods.** Competitor numbers are from their papers; no re-runs on our evaluation data (comparisons are directional). Palu quality is +1.29 PPL absolute on WikiText-2 Llama2-7B (≈+25 % relative). TurboQuant quality is from LongBench downstream scores (paper does not report PPL). CommVQ quality at 2-bit; degrades significantly at 1-bit. KVTC ratio is the stated peak. NexusQuant A10G rows: key-key scorer, all overhead included. A100 rows: new GPU-validated results (Sections 4.10–4.12).

Method	Ratio	Quality	Train-free	Notes
KVTC [15] (NVIDIA, ICLR 2026)	up to 20×	<1 pp	No (cal. req'd)	arXiv:2511.01815
Palu [2] (ICLR 2025)	11.4×	+1.29 PPL	No (cal. req'd)	Llama2-7B, WT-2
CommVQ [8] (Apple, ICML 2025)	≈8×	≈0 %	No (training)	2-bit config
TurboQuant [21] (Google, ICLR 2026)	≈4.6–5.1×	≈0 %	**Yes**	quant-only, no eviction
KIVI [11]	4–8×	<1 % PPL	Yes	2–4 bit
NexusQuant (short ctx, 500 tok)	**10.1**×	+0.90 %	**Yes**	Mistral-7B, A10G
NexusQuant (long ctx, 1664 tok)	**10.4**×	+0.14 %	**Yes**	Mistral-7B, A10G
NexusQuant (long ctx, 1664 tok)	**16.6**×	+0.82 %	**Yes**	Mistral-7B, A10G
NexusQuant (max, 1664 tok)	**32.7**×	+2.13 %	**Yes**	Mistral-7B, A10G
NexusQuant K3V2 (A100, 3544 tok)	**8.89**×	**+0.35** %	**Yes**	Mistral-7B, A100; 35 % evict
NexusQuant real scorer (A100, 764 tok)	≈**5**×†	**+0.66** %	**Yes**	Mistral-7B, A100; 80 % evict; †eviction-only ratio
NexusQuant K3V8 (L4, 1742 tok)	**2.6**×	**+0.71** %	**Yes**	Mistral-7B, L4; 35 % evict; near-lossless tier
NexusQuant K3V8 (L4, 1742 tok)	**4.2**×	**+1.03** %	**Yes**	Mistral-7B, L4; 60 % evict

and (3) quantization with aggressive eviction (8–33×, NIAH degraded, <1 % PPL at 16.6×) for memory-constrained deployments where "fits versus does not fit" dominates quality considerations. The cliff position depends on model, context length, and scorer; it must be validated per deployment configuration.

5.2 Comparison with Published Methods

Table 21 compares NexusQuant against published methods. Key observations:

vs. TurboQuant

TurboQuant (training-free, quantization-only) achieves 4.6–5.1× at near-zero quality loss [17], this ratio is *pure quantization with no eviction*. NexusQuant at long context achieves 10.4× at +0.14 % and 16.6× at +0.82 %, ratios that *include* 35–60 % token eviction on top of 2-bit E8 VQ. The comparison is not apples-to-apples in compression mechanism, but it is the relevant practitioner comparison: NexusQuant reaches 2–3.5× higher total compression at comparable or better quality. The underlying reason is that eviction and quantization are orthogonal compression axes: TurboQuant saturates around 5× (quantization minimum precision ceiling); NexusQuant extends beyond by adding a second, independent axis.

vs. KVTC: NexusQuant reaches 16.6× at <1 % completely training-free, and 32.7× (though at +2.1 % degradation) for aggressive configurations.

vs. Palu: +25 % relative) on Llama2-7B with calibration. NexusQuant achieves 10.4× at +0.14 % and 16.6× at +0.82 % on Mistral-7B without calibration.

evaluation metrics differ across methods; we have not run TurboQuant, KVTC, or Palu on our exact evaluation data. The numbers above should be understood as establishing competitive positioning in the training-free high-compression regime, not as controlled head-to-head experiments.

6 Limitations

We document all known limitations of the current work.

1. **Short-prefix cliff.** At ≈500-token prefixes, the safe operating boundary is 35 % eviction (10×). The eviction cliff is steep: 36 % eviction degrades by >2× compared to 35 %. Users must not exceed the safe boundary without verifying on their own distribution.

2. **Long-prefix catastrophic failure at high eviction.** At 2924-token prefixes with 60 %+ eviction, PPL degrades by >42 %. This is a fundamental capacity loss, not a scorer deficiency: we tested twelve importance scorers and all fail at this regime. The safe operating boundary at 2924 tokens is 35 % eviction (10×, +1.5 % PPL).

3. **Domain sensitivity.** Creative and narrative text is harder to compress than academic or technical text at low eviction rates, due to more uniform attention distributions. Users processing narrative content should reduce eviction rates by ≈10–15 % relative to technical-text defaults.

4. **Evaluation breadth.** Results are on Mistral-7B and Llama-3-8B only. The negative PPL result on Llama-3 is most likely an evaluation artifact (Section 4.4) and should not be generalized. We do not report LongBench task accuracy, RULER, or generation-quality evaluations. Full validation at 16K+ contexts has not been performed.

5. **Memory measurement only.** Compression ratios are from `torch.cuda.memory_allocated()` with all overhead. We do not report wall-clock latency or throughput; GPU kernels for compressed attention are not implemented.

6. **Competitor comparisons are directional.** We compare against published numbers for KVTC, TurboQuant, CommVQ, and Palu; we have not run their code on our evaluation data, so comparisons are not controlled experiments.

7. **Perplexity as proxy.** PPL correlates with downstream task quality but is not a perfect proxy. The QA evaluations (Section 4.7) confirm factual recall is preserved at 10×, but full LongBench evaluation remains for future work.

8. **PPL vs. downstream task metrics: a hard gap.** Section 5 documents the clearest failure of PPL as a quality indicator: K3V2 at 35 % eviction on Mistral-7B scores +0.82 % PPL ("safe") yet achieves only 40 % NIAH recall. Production validation on Gemma-2-2b-it at 4K context sharpens this: at 15 % eviction, PPL is +0.04 % (indistinguishable from quant-only) yet NIAH recall is zero despite 100 % needle token retention. PPL measures average next-token prediction, losing a few specific tokens barely moves the average over a 4K-token passage. NIAH measures worst-case factual recall, one evicted needle token destroys the answer. Any KV compression method that reports only perplexity is presenting an incomplete picture. The honest position for NexusQuant is this: the current pipeline preserves statistical language quality (as measured by PPL) but can damage specific fact recall in retrieval-sensitive applications, particularly for facts located early in the context. Users with retrieval workloads must evaluate NIAH or equivalent task-specific probes at their target configuration before deployment.

9. **GSM8K reasoning evaluation.** GSM8K reasoning evaluation was attempted but the baseline model achieved 0 % accuracy with the current prompt format, making the comparison invalid. Proper few-shot evaluation is future work.

10. **Gemma-2-2b-it under K3V2 shows a position-dependent NIAH scatter at relative position above ∼55 % of a 4K context.** Eight needle-position probes at absolute token indices {1800, 2200, 2600, 2800, 3000, 3200, 3400, 3600} on Gemma-2-2b-it K3V2 E8 quant-only produce a 4/8 pass rate with a scattered YES/NO pattern rather than a contiguous dead zone. Across six separate cycles we eliminated the following candidate mechanisms: alpha rescaling of amax, haystack content (Shakespearean and legal text substitutions), needle identity (alternative needle sentence), quantizer choice (E8 vs uniform vs Lloyd-Max), K precision (K3V2 vs K4V2 gives 4/8 vs 5/8), protect_boundary (0 vs 2), absolute token position (verified via a 6K context probe that was sliding-window-safe), `attn_logit_softcapping` (disabled via monkey-patch on all 26 attention layers), and the RoPE phase at the failure indices. A cross-model control on Qwen2.5-7B-Instruct K3V2 at the

same eight indices produces a 7/8 pass rate, ruling out both a general 4-KV-head K3V2 limitation and an information-theoretic precision bound (Qwen has fewer bits per token per head than Gemma-2-2b yet performs better). The remaining explanation is an interaction specific to Gemma-2-2b between its hybrid sliding-window attention, its 2B parameter capacity, and K quantization. We recommend Gemma-2-9b or larger on the Gemma-2 family, or avoiding K3V2 on Gemma-2-2b for long-context needle retrieval workloads.

7 Conclusion

NexusQuant achieves 13.7× KV cache compression on Mistral-7B with 8/8 NIAH recall (versus 7/8 for uncompressed FP16), requiring zero calibration. The method combines E8 lattice vector quantization with Hadamard rotation and entropy coding: at 2-bit quantization, E8 lattice indices cluster into only 5 unique coordinate values (Shannon entropy 1.32 bits/symbol), enabling `zstd` to nearly triple the raw compression ratio. This structural clustering is a property of lattice quantization absent in scalar approaches.

FP16 baseline tests across Mistral-7B, Qwen2.5-7B, and Gemma-2-2b reveal position-dependent retrieval dead zones in every model, confounding prior NIAH evaluations that attributed all failures to compression. Baseline-adjusted, K3V2 compression damage on Mistral is zero; on Gemma-2-2b it is one position out of five. On four independent instances across two models, Hadamard+E8 quantization rescues positions where the uncompressed model fails, consistent with a stochastic-resonance mechanism where structured quantization noise perturbs borderline attention peaks above the retrieval threshold.

Broader Impact

Positive impacts

KV cache compression lowers the hardware barrier for long-context LLM inference. At 10–16× compression, a model that previously required 80 GB of GPU memory for 128K-token contexts can run on a 24 GB consumer GPU, or on the same hardware at 1.3–2.5M-token contexts. This has several downstream effects. First, it democratizes long-context deployment: research groups and companies without access to multi-GPU clusters can run long-context inference workloads that are currently cost-prohibitive. Second, it reduces inference cost directly: more aggressive KV compression means larger effective batch sizes in memory-bound serving scenarios, reducing cost per query proportionally. Third, it enables on-device deployment of longer contexts: mobile or edge devices with 8–16 GB of available memory could handle contexts that today require server-side infrastructure.

Negative impacts

The same cost reduction that makes long-context inference more accessible also lowers barriers for misuse. Cheaper inference makes it easier to deploy large-scale language models with less oversight, including for harmful content generation, disinformation, or automated manipulation at scale. Compressed KV caches can introduce subtle quality degradation that is difficult to detect in production: our domain sensitivity results (Section 4.8) show that creative and narrative text is systematically harder to compress than academic text, and PPL degradation at high compression ratios may manifest as factual errors or incoherent reasoning that are not easily caught without task-specific evaluation. Additionally, token eviction may disproportionately drop tokens corresponding to minority-language passages, rare-topic sentences, or low-frequency vocabulary, because attention-based importance scores are calibrated on the majority distribution. A document mixing common and rare languages may have its rare-language tokens systematically evicted.

Mitigations

We provide quality presets (`quality="lossless"` at 10x, `quality="balanced"` at 16x, `quality="aggressive"` at 32x) with documented PPL degradation ranges from our GPU experiments. We strongly recommend domain-specific sensitivity testing before production deployment: the operating point safe for academic text (35 % eviction, +0.14 % PPL) is not safe for creative text (+2.48 % PPL at the same setting). The catastrophic failure mode at 60 %+ eviction with 3K-token prefixes is documented in the limitations section and surfaced as a hard warning in the API. We recommend that users deploying NexusQuant in multilingual or low-resource-language settings evaluate compression quality separately on their target language distribution before relying on default presets.

References

[1] Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, et al. PyramidKV: Dynamic KV cache compression based on pyramidal information funneling. *arXiv preprint arXiv:2406.02069*, 2024.

[2] Chi-Chih Chang, Wei-Cheng Lin, Chien-Yu Lin, Natalia Frumkin, Diana Marculescu, Sergi Abadal, et al. Palu: KV-cache compression with low-rank projection. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.21118.

[3] Jerry Chee, Yaohui Cai, Volodymyr Kuleshov, and Christopher De Sa. QuIP#: Even better LLM quantization with Hadamard incoherence and lattice codebooks. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.

[4] Insu Han, Praneeth Kacham, Amin Karbasi, Vahab Mirrokni, and Amir Zandieh. PolarQuant: Quantizing KV caches with polar transformation, 2025. arXiv:2502.02617.

[5] Jitai Hao, Qiang Huang, Yaowei Wang, Min Zhang, and Jun Yu. DeltaKV: Semantic residual compression for KV caches. *arXiv preprint arXiv:2602.08005*, 2026.

[6] Coleman Hooper, Sehoon Kim, Hiva Mohammadi, Kurt Keutzer, Michael W. Mahoney, and Amir Gholami. KVQuant: Towards 10 million context length LLM inference with KV cache quantization. *arXiv preprint arXiv:2401.18079*, 2024.

[7] Hao Kang, Qingru Zhang, Souvik Kundu, Geonhwa Jeong, Zaoxing Liu, Tushar Krishna, and Mosharaf Chowdhury. GEAR: An efficient KV cache compression recipe for near-lossless generative inference of LLM. *arXiv preprint arXiv:2403.05527*, 2024.

[8] Junyan Li, Yang Zhang, Muhammad Yusuf Hassan, et al. CommVQ: Commutative vector quantization for KV cache compression. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. arXiv:2506.18879.

[9] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Vempati, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[10] Yuhan Liu, Hanchen Li, Yihua Cheng, Siddhant Ray, Yuyang Huang, Qizheng Xu, Kuntai Chen, Jiayi Guo, Weijia Lu, Bhaskar Krishnamachari, et al. CacheGen: KV cache compression and streaming for fast large language model serving. In *Proceedings of ACM SIGCOMM*, 2024.

[11] Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A tuning-free asymmetric 2-bit quantization for KV cache. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2024.

[12] Junlin Mu, Hantao Huang, Jihang Zhang, et al. SALS: Spectral analysis for low-rank KV cache compression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2510.24273.

[13] Kaleem Ullah Qasim, Jiashu Zhang, Muhammad Kafeel Shaheen, Razan Alharith, and Heying Zhang. The residual stream is all you need: On the redundancy of the KV cache in transformer inference. *arXiv preprint arXiv:2603.19664*, 2026.

[14] Semyon Savkin, Eitan Porat, Or Ordentlich, and Yury Polyanskiy. NestQuant: Nested lattice quantization for neural network compression. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.09720.

[15] Konrad Staniszewski, Adrian Lancucki, et al. KVTC: KV cache transform coding for compact storage in LLM inference. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. arXiv:2511.01815.

[16] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2023.

[17] TheTom and contributors. TurboQuant+: Asymmetric KV precision via attention-noise analysis. `https://github.com/TheTom/turboquant_plus`, 2026. GitHub: TheTom/turboquant_plus.

[18] Tycho F. A. van der Ouderaa, Mart van Baalen, Paul Whatmough, and Markus Nagel. Leech lattice vector quantization for efficient LLM compression. *arXiv preprint arXiv:2603.11021*, 2026.

[19] Maryna S. Viazovska. The sphere packing problem in dimension 8. *Annals of Mathematics*, 185(3):991–1015, 2017.

[20] Yixuan Wang, Qingyu Shi, Jiayu Zhou, Dianbo Liu, Ziwei He, and Zhouhan Lin. VQKV: High-fidelity and high-ratio cache compression via vector-quantization. *arXiv preprint arXiv:2603.16435*, 2026.

[21] Amir Zandieh, Majid Daliri, Mehdi Hadian, and Vahab Mirrokni. TurboQuant: Online vector quantization with near-optimal distortion rate. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. arXiv:2504.19874.

[22] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. H_2O: Heavy-hitter oracle for efficient generative inference of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[23] Ciaran Zhou et al. EliteKV: Scalable KV cache compression via RoPE frequency selection and joint low-rank projection. *arXiv preprint arXiv:2503.01586*, 2025.

Appendix

A E8 Lattice: Algorithm and Relaxed Parity

Given $v \in \mathbb{R}^8$, compute: (1) round v to nearest integer vector $z_{\mathbb{Z}}$ and adjust to even sum; (2) round $v - \frac{1}{2}\mathbf{1}$ to nearest integer vector and adjust to even sum, then add $\frac{1}{2}\mathbf{1}$ to get $z_{1/2}$; (3) return $\arg\min_{z \in \{z_{\mathbb{Z}}, z_{1/2}\}} \|v - z\|^2$.

Relaxed parity justification

Strict E_8 requires $\sum_i x_i \equiv 0 \pmod 2$ for *both* cosets. Empirically on KV cache data, relaxing this for the half-integer coset reduces MSE by 0.3–0.4 % consistently across layers and heads on Mistral-7B. The mechanism is that Hadamard-rotated KV vectors are approximately sub-Gaussian, so the optimal codebook has slightly higher density near the origin than strict E_8 provides. The relaxed lattice ($\mathbb{Z}^8_{\text{even}} \cup (\mathbb{Z} + \frac{1}{2})^8$) achieves this without any training.

B Negative-PPL Investigation Details

For Llama-3-8B at 1494-token prefix, we ran five controlled ablations to identify the source of negative PPL:

1. **Mask-only (no eviction)** masked evicted positions to $-\infty$ but kept the vectors: PPL$\Delta = -2.24\,\%$. This isolates the masking effect.
2. **Zero-fill (no mask)**: set evicted vectors to zero without masking: PPL$\Delta = +5100\,\%$. Catastrophic.
3. **Random eviction**: randomly selected positions (not importance-based): PPL$\Delta = -1.60\,\%$ (average). The improvement is not scorer-dependent.
4. **Quant-only (no eviction)**: PPL$\Delta = +4.37\,\%$ at 2-bit.
5. **Evict + quant (importance-based)**: PPL$\Delta = -0.47\,\%$.

Conclusion: the $-\infty$ masking causes attention renormalization that sharpens the model's distribution over the retained suffix. This is an evaluation artifact: it reflects a shorter effective context, not better model quality. The effect is GQA-structure-dependent (Llama-3-8B uses 4:1 GQA; Mistral-7B does not), which explains why it appears here and not on Mistral. We report these results for completeness and reproducibility but do not claim them as evidence that NexusQuant improves generation quality.

C Failed Approaches (12 Experiments)

We tested the following approaches and found they fail or offer no benefit. We document them to prevent future duplication.

1. **L2-norm importance scorer**: +207 % PPL. L2 norm $\neq$ attention importance; completely wrong token set retained.
2. **Token zeroing for eviction** (without $-\infty$ mask): catastrophic. $\text{softmax}(Q@\mathbf{0}) = \text{softmax}(\mathbf{0}) = \mathbf{1}/T$; evicted tokens steal attention.

3. **Strict E8 parity fix**: +0.3–0.4 % worse. Relaxed parity is an empirical improvement, not a bug to fix.

4. **QJL 1-bit residual correction**: no improvement; distributes residual noise to well-quantized dimensions. This finding is triple-confirmed: (1) our own GPU experiments show no quality gain from QJL residual on top of E8 quantization; (2) the TurboQuant community independently found that *disabling* the QJL residual in TurboQuant improves quality by +21.5 pp (MidasMining, vllm#38479); (3) SpectralQuant theory explains the mechanism, the JL projection injects noise into 125 of 128 dimensions that were already well-quantized, corrupting good dimensions to marginally improve the 3 worst ones. Residual correction after lattice VQ is counterproductive.

5. **Cross-layer KV delta**: 0 bits saved. Layers project to orthogonal subspaces.

6. **Per-head independent eviction at high rates**: degrades faster than aggregated eviction.

7. **Variable-rate E8 (per-group water-filling)**: +3.67 % at matched rate (3.3× worse). Energy ≠ importance.

8. **3-bit + eviction (combined)**: no compression gain over 2-bit; entropy coding more efficient at 2-bit.

9. **Mixed K2V3 (2-bit keys, 3-bit values)**: marginal and complex. Note: the *reverse* asymmetry, K3V2 (3-bit keys, 2-bit values), is *not* a failure; it is validated and reported in Section 4.12. The distinction matters: keys warrant more bits than values because key quantization noise corrupts all attention weights globally, whereas value noise only scales individual output contributions.

10. **PCA dimension reduction**: worse than Hadamard at matched bitrate.

11. **Observation-window scaling**: no effect on quality; scoring is robust to window size.

12. **Token merging**: introduces decompression overhead that erases compression gains.